

G.V ACHARYA INSTITUTE OF ENGINEERING
& TECHNOLOGY

Veeracharya Technical Education Campus, Opp. Shelu Railway Station, (E), 2nd Station After Badlapur Pin - 410 101

Student Name: Manan Vernekar Subject Java Date _____

Start From here _____

* Package - Collection of files

• Java consists of Classes & Objects.

* Class Name = File Name.

• Functions do specific task.

* // used to comment.

* ^(tab) cout = to print = system.out.println(); ↙ to add line space.

* Methods = functions written in Classes.

* ; used to show line ends.

* Variables - to store data in different forms.

Data Types

Primitive

Non Primitive

Memories (byte(s))

Byte - 1

Short - 2

Int - 4

Long - 8

Float - 4

Double - 8

Char - 2

Boolean - 1

String

Arrays.

- Use $\Rightarrow$ long number = 12345678900L
↑
(To mention long number)

- (F $\rightarrow$ for float)

* String Functions -

- cout(Cname.length()); $\rightarrow$ to get number of chars.

- * Byte stores numbers from (-128 to 127)

- * Primitive have fixed size.

- * Non-primitive have undefined size.

- concatenate = string1 + string2 = string3.

- .charAt() - for checking character ^{at mentioned} position.

- .replace() - to replace character with other character.

- .substring() - to get substring from the string mentioned.

- * Arrays - store same data types.

- to declare $\rightarrow$ eg

```
int[] marks = new int[3]
marks[0] = 23;
marks[1] = 24;
marks[2] = 30;
```

- to access data $\rightarrow$ eg.

cout(marks[0]);

- * Ctrl + / - to comment selected part of code.

- * Java puts unentered/empty variables as null/0, by default.

- .length - to get total number of variables in Array.

- * In java classes name are mostly starting with capital letter.

- .sort - Use to sort the variables in array to ascending order.

- * import - used to import packages/packages.

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- * If we know the values in an array/string we can directly define them in the code without using 'new' keyword.

eg- `int[] roll_no = {13, 16, 25, 49};`

- * 2D Arrays.

eg-

`int[][] marks = {{25, 29, 30}, {10, 30, 23}};`

To call - `solut(marks[i][j]);`

- * Casting.

1) Implicit Casting - can store values like 'int' in 'doubles' as 'int' has less size than 'double'.

2) Explicit Casting - to store values of high size in low size we use explicit casting, by losing some value in the casting process.

eg- To store double in int.

`int fprice = price + (int)18.0;` ← it will be removed.

- * Constants -

To declare a constant eg.

final float PI = 3.14F;

so it doesn't modify later.

- * Use capital letters for final variables like above.

* Operators

- 1) Arithmetic Operators - normal mathematics operators like +, -, etc.
- 2) Assignment Operators - to assign values like =.
- 3) Unary Operator = eg- $++num$ = will add 1 in the number prior if, $num++$ = will add the number with 1 afterwards.
- 4) Decrement Operator = $--num$
num will reduce by 1, prior.
 - $num--$ = will reduce the number afterwards.

* Math Class - includes maths functions likes max, min, etc.
• random - range $0 \rightarrow 1$.

* Scanner Class - to take user input.

- $sc.nextInt()$; - to take 'int' as input.
- $sc.next()$; - as String, but only single word/token.
- $sc.nextLine()$; - for complete line of String.

* Comparison Operators -

- $a == b$ $\rightarrow$ equal to operator.
- $a != b$ $\rightarrow$ not equal to.
- $a < b$ $\rightarrow$ greater than
- etc.

* Conditional Statements.

$\left(\begin{array}{l} \bullet \text{if ()} \{ \\ \quad \text{---} \} \\ \text{else} \\ \quad \{ \text{---} \} \end{array} \right)$ and $\left(\text{else if ()} \{ \text{---} \} \right)$

* Logical Operators

- $\&\&$ - and operators
- $\|\|$ - or operators
- $!$ - negative / not operator.

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* Switch - This conditional statement is used to switch cases from 'case 1' to 'case n' or 'case default' and execute.
→ switch () { }

case 1:

case 2:

case n:

* Break;^b - used to break the switch cases.

* Loops -

1) for () { } - C initial; condition; update { }

2) while () { } - use ^{loop} until the while condition is ~~is~~ satisfied.
_{initial condition.}

* We can't redeclare same name two variables, but in for loop the variable name is isolated.

* In while loop ~~condi~~ 'update' statement is written outside the ().

3) do { } while () * while condition is ~~is~~ satisfied.
→ do a task until

* continue; - used to skip the condition that's satisfied and continue loop ahead.

* Exception Handling - Exceptions can be handled, but Errors can't.

• Exceptions

• OutOfBounds - is the variable which is out of the ideal length defined.

• try {} - to get the exception.

• catch {} - to store the exception.

↳ to do after catch.

* Exceptions can be handled and the remaining program works fine.

* Functions - are the operations performed or tasks done.

* Methods - are the small tasks performed, which comes under a single 'Class'.

• Repeatable tasks can be stored in 'methods', inside a 'Class' together.

eg- public static void printJava() {
 sout("Hello Java");
}

To call -

printJava();
printJava();

function used.

eg- public static void printName(String name) {
 sout(name);
}

To call -

printName("Manan");

* Package

↳ public class Main

↳ public static void main (String[] args).

similarly eg,

eg>> public static void printHello (String name).